CAREERS360

PRACTICE Series

NEET UG

Botany Important Exceptions: NCERT-Based Quick Revision PDF

Download FREE PDF

CAREERS360 PRESENTS

Important Exceptions of NEET Botany

A chapter-wise, NCERT-based guide to every rule-breaking organism and physiological exception that NEET aspirants must know

Based on the Official NEET (UG) 2026 Syllabus
issued by the National Medical Commission (NMC)

www.careers360.com

careers360

About This Ebook

NEET (UG) Botany is a subject built on general rules — every plant group, tissue system and physiological pathway seems to follow a clean, predictable pattern. But NTA loves to test the plants and mechanisms that break these very rules. Questions like 'which seed plant has motile sperm', 'which pathway fixes CO2 at night' or 'which fruit forms without fertilisation' appear year after year, and are exactly the kind of high-yield, easy-to-miss facts that separate a good score from a great one.

This ebook, Important Exceptions of NEET Botany, has been compiled by Careers360 strictly on the basis of the official NEET (UG) 2026 syllabus notified by the National Medical Commission (NMC), and follows the NCERT textbook sequence chapter by chapter — from Diversity in the Living World all the way to Ecology and Environment.

For every exception covered, you will find four things laid out clearly:

- General Rule — the standard pattern most organisms/processes follow
- The Exception — the organism or process that breaks this rule
- Why (Reasoning) — the biological logic behind the exception
- NCERT / Classic Example — the exact example you should recall in the exam

Use this ebook as a rapid-revision companion in the final weeks before your exam — read it chapter-wise alongside your NCERT textbook, or skim through it a day before the exam as a quick-recall sheet for these high-value, frequently-tested facts.

All the best for NEET (UG) 2026!

careers360

Table of Contents

Click any chapter title below to jump directly to it.

Chapter 1: Diversity in the Living World — Plant Classification

Chapter 2: Structural Organisation in Plants

Chapter 3: Cell Structure and Function — The Plant Cell

Chapter 4: Photosynthesis in Higher Plants

Chapter 5: Respiration in Plants

Chapter 6: Plant Growth and Development

Chapter 7: Sexual Reproduction in Flowering Plants

Chapter 8: Genetics and Evolution in Plants

Chapter 9: Plants and Microbes in Human Welfare

Chapter 10: Biotechnology and Its Applications in Plants

Chapter 11: Ecology and Environment — Plant Interactions

Other Useful Resources

careers360

Chapter 1: Diversity in the Living World — Plant Classification

Plant (and plant-adjacent) classification is built on general rules of cell structure, pigments and reproduction. NEET regularly tests the organisms that break these rules — memorise these carefully.

1. A 'fungus' with a cellulose cell wall

General Rule: Fungi are defined, among other things, by a cell wall made of chitin.

The Exception: Oomycetes (water moulds), historically studied alongside fungi, have a cell wall made of cellulose instead of chitin.

Why (Reasoning): This fundamental difference in cell wall chemistry is one of the key reasons Oomycetes are no longer classified within Kingdom Fungi in modern taxonomy, despite their fungus-like filamentous growth.

NCERT / Classic Example: Pythium and Albugo (the causal organism of white rust) are the oomycetes cited in this context.

2. A bacterium with no cell wall at all

General Rule: Bacteria (Kingdom Monera) are generally characterised by a rigid cell wall that gives them a definite shape.

The Exception: Mycoplasma completely lacks a cell wall, making it the smallest living cell and giving it a flexible, variable shape.

Why (Reasoning): Without a rigid wall, Mycoplasma can squeeze through filters that trap ordinary bacteria and can survive without oxygen, which is why it is often highlighted as an exception in the Monera classification.

NCERT / Classic Example: Mycoplasma — can even survive without oxygen, unlike most bacteria.

3. Two organisms living as one

General Rule: Each organism studied under the five-kingdom classification is treated as a single, independent living entity.

The Exception: A lichen is not one organism but a symbiotic association between a fungus (mycobiont) and a photosynthetic partner — an alga or cyanobacterium (phycobiont) — living so intimately together that they were long mistaken for a single species.

Why (Reasoning): The fungus provides shelter/absorbs water and minerals while the alga/cyanobacterium provides food via photosynthesis — a mutualistic partnership tight enough to produce a completely new, composite growth form.

NCERT / Classic Example: Lichens as indicators of air pollution (they are highly sensitive to SO2).

4. A seed plant with swimming sperm

General Rule: Seed plants (gymnosperms and angiosperms) deliver male gametes to the egg through a pollen tube, so their sperm do not need flagella to swim.

The Exception: Cycas, a gymnosperm, still produces motile, flagellated (multiciliate) sperm — a primitive holdover from its fern-like ancestors.

Why (Reasoning): This is considered a transitional feature linking gymnosperms to the free-sporing, water-dependent fertilisation of pteridophytes.

NCERT / Classic Example: Cycas — one of the very few seed plants with motile sperm.

Careers360

5. Pteridophytes that produce two kinds of spores

General Rule: Most pteridophytes are homosporous — they produce only one type of spore, which develops into a bisexual gametophyte.

The Exception: Selaginella and Salvinia are heterosporous, producing distinct microspores (male) and megaspores (female) — a condition otherwise typical of seed plants.

Why (Reasoning): Heterospory in these pteridophytes is considered evolutionarily significant, as it foreshadows the seed habit seen in gymnosperms and angiosperms.

NCERT / Classic Example: Selaginella and Salvinia — explicitly named in NCERT as the heterosporous pteridophytes.

6. A 'plant' with no chlorophyll

General Rule: Plants are, by definition, autotrophic — they synthesise their own food using chlorophyll.

The Exception: Cuscuta (dodder) is a total stem parasite that has lost its chlorophyll and leaves entirely, deriving all its nutrition directly from a host plant.

Why (Reasoning): Having no photosynthetic pigment of its own, Cuscuta survives purely as a heterotroph, attaching haustoria into the host's vascular tissue to withdraw water, minerals and food.

NCERT / Classic Example: Cuscuta — the classic example of a total stem parasite in NCERT.

careers360

Chapter 2: Structural Organisation in Plants

Roots, stems and leaves usually stick to their textbook jobs — underground anchorage, aerial support, and photosynthesis respectively. A set of striking modifications breaks this pattern.

1. Roots that breathe air

General Rule: Roots are underground organs specialised for anchorage and absorption of water and minerals.

The Exception: In mangrove plants growing in waterlogged, oxygen-poor swamps, specialised roots called pneumatophores grow vertically upward, emerging above the mud/water to absorb atmospheric oxygen through pores called pneumatothodes.

Why (Reasoning): Since the soil around mangroves is saline and anaerobic, normal underground respiration is not possible, so these roots evolved a purely respiratory function instead of an absorptive one.

NCERT / Classic Example: Rhizophora (mangrove) — pneumatophores or 'breathing roots'.

2. Roots that hang in the air

General Rule: Roots normally grow into the soil, where they are shielded from desiccation.

The Exception: Epiphytic orchids growing on tree trunks develop aerial roots covered with a spongy tissue called velamen, which absorbs moisture directly from the humid air.

Why (Reasoning): Because epiphytes have no access to soil, this modification lets them survive entirely on atmospheric moisture and nutrients washed down the host tree's bark.

NCERT / Classic Example: Vanda and other epiphytic orchids — velamen roots.

3. A stem that looks and works like a leaf

General Rule: In a typical plant, the stem provides support while the leaves are the main photosynthetic organs.

The Exception: In cacti such as Opuntia, the leaves are reduced to spines (to minimise water loss), and the flattened, green, fleshy stem (a phylloclade) takes over photosynthesis entirely.

Why (Reasoning): This division of labour — spines for defence/reduced transpiration and stem for photosynthesis — is a classic desert (xerophytic) adaptation.

NCERT / Classic Example: Opuntia — phylloclade performing photosynthesis instead of leaves.

4. Water-conducting tissue with no vessels

General Rule: In most flowering plants (angiosperms), xylem contains vessels — wide, efficient tubes for water conduction.

The Exception: Gymnosperms lack vessels in their xylem altogether; water is conducted solely through narrower tracheids.

Why (Reasoning): The absence of vessels makes gymnosperm wood structurally different ('softwood') and is considered a more primitive condition compared to the vessel-bearing xylem of angiosperms.

NCERT / Classic Example: Pinus and other gymnosperms — xylem without vessels.

5. Food-conducting cells with no companions

Careers360

General Rule: In angiosperm phloem, sieve tube elements are always accompanied by companion cells that manage their metabolic activities.

The Exception: Gymnosperms lack both sieve tube elements and companion cells; instead, their phloem contains sieve cells associated with albuminous cells, which perform a similar supportive role.

Why (Reasoning): This is another structural marker that distinguishes the more primitive vascular tissue organisation of gymnosperms from that of angiosperms.

NCERT / Classic Example: Gymnosperm phloem — sieve cells and albuminous cells instead of sieve tubes and companion cells.

careers360

Chapter 3: Cell Structure and Function — The Plant Cell

Even at the level of a single plant cell, a few components refuse to follow the 'usual' pattern.

1. A living cell with (almost) no cytoplasm of its own

General Rule: A living, functioning cell is expected to retain its own nucleus to direct and regulate its metabolic activities.

The Exception: Mature sieve tube elements (the conducting cells of phloem) lose their nucleus at maturity, yet remain alive — their functions are regulated by an adjacent, nucleated companion cell connected to them by numerous plasmodesmata.

Why (Reasoning): This is a unique division of labour: the companion cell essentially acts as the 'control centre' for its partner sieve tube element, which is otherwise reduced to a simple conducting channel.

NCERT / Classic Example: Sieve tube elements and companion cells in angiosperm phloem.

2. An organelle that takes up almost the whole cell

General Rule: In a typical animal cell, vacuoles are small and relatively insignificant in volume.

The Exception: A mature plant cell often has a single, large central vacuole that can occupy up to 90% of the cell's volume, pushing the cytoplasm to a thin peripheral layer.

Why (Reasoning): This large vacuole maintains turgor pressure (rigidity) in plant cells, stores various metabolites/pigments, and helps the cell grow rapidly by cheaply increasing volume with water rather than costly cytoplasm.

NCERT / Classic Example: Mature parenchyma cells of any flowering plant.

3. One organelle turning into another

General Rule: Organelles are generally considered to have a fixed identity and function throughout the life of the cell.

The Exception: Plastids can interconvert — a colourless leucoplast (starch-storing) can transform into a pigmented chromoplast, and chloroplasts themselves can turn into chromoplasts as a fruit or root matures.

Why (Reasoning): This flexibility allows plant tissues to change colour and store different materials as they develop, such as when a green, starchy structure converts into a brightly coloured, sugar-storing one.

NCERT / Classic Example: The colour change of a ripening tomato, or the orange colour of a carrot root, both reflect chromoplast formation.

careers360

Chapter 4: Photosynthesis in Higher Plants

Not every plant photosynthesises the 'standard' C3 way — and not every photosynthetic organism even produces oxygen.

1. A CO_2-fixing pathway with an extra first step

General Rule: In the standard (C3) pathway, atmospheric CO_2 is fixed directly into a 3-carbon compound by the enzyme RuBisCO in the Calvin cycle.

The Exception: C4 plants first fix CO_2 into a 4-carbon compound using the enzyme PEP carboxylase in mesophyll cells, before the CO_2 is released and re-fixed by RuBisCO in specialised bundle-sheath cells (Kranz anatomy).

Why (Reasoning): This two-step CO_2-concentrating mechanism minimises photorespiration and water loss, making C4 plants far more efficient in hot, dry, high-light environments than ordinary C3 plants.

NCERT / Classic Example: Maize (Zea mays), sugarcane and sorghum are classic C4 plants.

2. Fixing carbon at night, using it in the day

General Rule: In most plants, both the initial CO_2 fixation and the Calvin cycle occur during the daytime, when stomata are open for gas exchange.

The Exception: In CAM (Crassulacean Acid Metabolism) plants, stomata open only at night to fix CO_2 into organic acids, which are stored in vacuoles and only used in the Calvin cycle the following day, with stomata closed.

Why (Reasoning): This temporal separation lets desert succulents drastically cut water loss, since they never need to keep stomata open during the hot, dry daytime.

NCERT / Classic Example: Opuntia and other succulents show CAM photosynthesis.

3. An enzyme that works against itself

General Rule: RuBisCO's primary job is to act as a carboxylase, fixing CO_2 onto RuBP to drive the Calvin cycle.

The Exception: RuBisCO can also act as an oxygenase, combining RuBP with O_2 instead of CO_2 — a wasteful process called photorespiration that consumes energy without producing any sugar or net ATP.

Why (Reasoning): Photorespiration becomes prominent in C3 plants when CO_2 levels drop and O_2 levels rise inside the leaf (e.g., on a hot day with stomata partly closed), representing a built-in inefficiency of the C3 pathway that C4 plants largely avoid.

NCERT / Classic Example: C3 plants like wheat and rice are more prone to photorespiration than C4 plants.

4. Photosynthesis that releases no oxygen

General Rule: Photosynthesis is generally understood to split water and release oxygen as a by-product.

The Exception: Certain bacteria, such as purple and green sulphur bacteria, carry out anoxygenic photosynthesis, using hydrogen sulphide (H_2S) instead of water as the electron donor, and releasing sulphur instead of oxygen.

Why (Reasoning): This observation, central to Van Niel's hypothesis, proved that the oxygen released in ordinary plant photosynthesis comes from water, not from CO_2 — a landmark deduction based on exactly this kind of exception.

NCERT / Classic Example: Purple and green sulphur bacteria — anoxygenic photosynthesis.

Careers360

Chapter 5: Respiration in Plants

Respiration is not always a clean, complete oxidation of glucose to CO2 and water — plants (and their pathways) show some notable deviations.

1. Respiration without oxygen

General Rule: Cellular respiration is normally an aerobic process, in which glucose is completely oxidised to CO2 and water, yielding a large amount of ATP.

The Exception: In the absence of oxygen, cells (and organisms like yeast) switch to anaerobic fermentation, incompletely breaking down glucose into ethanol and CO2 (or lactic acid), releasing far less energy.

Why (Reasoning): Because glucose is not fully oxidised, fermentation yields a net of only 2 ATP per glucose molecule, compared to the far higher ATP yield of aerobic respiration — a major inefficiency tolerated only when oxygen is unavailable.

NCERT / Classic Example: Yeast (Saccharomyces) undergoing alcoholic fermentation; muscle cells undergoing lactic acid fermentation during oxygen debt.

2. A ratio that isn't always 1

General Rule: The respiratory quotient (RQ), the ratio of CO2 released to O2 consumed, is often assumed to be a fixed value.

The Exception: RQ actually varies with the respiratory substrate — it equals 1 for carbohydrates, is less than 1 (around 0.7) for fats and proteins, and can be greater than 1 for organic acids.

Why (Reasoning): This variation reflects the differing oxygen requirements needed to fully oxidise substrates with different proportions of carbon, hydrogen and oxygen in their molecular structure.

NCERT / Classic Example: RQ of about 0.7 when fat is the respiratory substrate, versus RQ = 1 for carbohydrate substrates.

3. A 'breakdown' pathway that also builds things up

General Rule: Respiration is generally thought of as a catabolic (breakdown) pathway whose sole job is to release energy from food.

The Exception: The respiratory pathway is amphibolic — many of its intermediates (such as those from glycolysis and the Krebs cycle) are diverted to build fatty acids, amino acids and other biomolecules, rather than being oxidised further.

Why (Reasoning): This dual (catabolic + anabolic) nature means respiration is not a one-way energy-releasing street but also a hub that supplies raw material for the cell's biosynthetic needs.

NCERT / Classic Example: Krebs cycle intermediates being withdrawn for amino acid synthesis.

careers360

Chapter 6: Plant Growth and Development

Growth regulation in plants includes some genuinely unusual hormones and dose-dependent effects.

1. Growth that never really stops

General Rule: Most animals show determinate (closed) growth — they grow to a fixed adult size and then stop.

The Exception: Plants show indeterminate (open) growth: because of the continuous activity of meristems (apical and lateral), a plant can, in principle, keep adding new roots, stems, leaves and branches throughout its life.

Why (Reasoning): This is possible because meristematic cells retain the ability to divide indefinitely, unlike most differentiated animal tissues.

NCERT / Classic Example: Continued height and girth increase in perennial trees throughout their lifespan.

2. A plant hormone that is a gas

General Rule: Plant hormones (like auxins, gibberellins and cytokinins) are generally organic molecules transported through vascular tissue in liquid form.

The Exception: Ethylene is a simple gaseous hydrocarbon hormone that diffuses through plant tissues and the surrounding air.

Why (Reasoning): Being volatile, ethylene can influence not just the plant that produces it but also fruit and flowers on neighbouring plants exposed to the gas.

NCERT / Classic Example: Ethylene is used commercially to promote fruit ripening (e.g., in mangoes and bananas) and to trigger flowering in some plants (e.g., pineapple).

3. The same hormone, opposite effects

General Rule: A given plant hormone at a given concentration is generally assumed to have a consistent, predictable effect on plant growth.

The Exception: Auxin promotes cell elongation and apical dominance in shoots at typical concentrations, but the same hormone (or slightly higher concentrations) inhibits root growth.

Why (Reasoning): Roots and shoots have very different sensitivities to auxin, so what is a growth-promoting concentration for the shoot can be a growth-inhibiting concentration for the root — a key exception to the 'one hormone, one effect' expectation.

NCERT / Classic Example: Removal of the shoot tip (apical bud) — which is the main source of auxin — releases lateral buds from apical dominance, letting them grow out.

4. Flowering that needs a cold snap first

General Rule: Flowering in most plants is regulated primarily by day length (photoperiod).

The Exception: In vernalisation, certain plants (many biennials/winter annuals) must first experience a period of low temperature before they are able to flower, regardless of day length.

Why (Reasoning): This cold requirement prevents premature flowering before winter and ensures that flowering (and seed production) occurs in a favourable season the following year.

NCERT / Classic Example: Winter varieties of wheat require vernalisation before they will flower.

5. Seeds that need light — not darkness — to germinate

Careers360

General Rule: Seeds are generally assumed to germinate once conditions of adequate water, oxygen and suitable temperature are met, irrespective of light.

The Exception: Some seeds are positively photoblastic and germinate much better (or only) when exposed to light, while others are negatively photoblastic and germinate better in the dark.

Why (Reasoning): This light-sensitivity, mediated by the phytochrome pigment system, is an adaptation that helps a seed 'sense' whether it is buried too deep or shaded out (unfavourable) versus near the surface with good light exposure (favourable).

NCERT / Classic Example: Lettuce seeds are a classic example of positively photoblastic seeds, requiring light to germinate well.

careers360

Chapter 7: Sexual Reproduction in Flowering Plants

Flowering plants show some remarkable reproductive shortcuts and specialisations that deviate from 'typical' sexual reproduction.

1. A seed made without fertilisation

General Rule: Sexual reproduction, by definition, requires the fusion of male and female gametes (fertilisation) to produce an embryo and, eventually, a seed.

The Exception: In apomixis, a seed is produced asexually — the embryo develops without the fusion of gametes, yet the end result mimics normal sexual reproduction in appearance.

Why (Reasoning): Apomixis is exploited commercially because it allows a hybrid plant's superior traits to be maintained across generations without the genetic reshuffling that normally comes with sexual reproduction — of great interest for hybrid seed technology.

NCERT / Classic Example: Certain Citrus and Asteraceae members show apomixis.

2. A fruit that develops without fertilisation

General Rule: Fruit development is normally triggered by fertilisation of the ovule following pollination.

The Exception: In parthenocarpy, fruits develop without fertilisation ever occurring, and are consequently seedless.

Why (Reasoning): This trait is deliberately exploited in horticulture to produce popular seedless fruit varieties.

NCERT / Classic Example: Seedless varieties of banana are a well-known parthenocarpic fruit.

3. More than one baby per seed

General Rule: A typical angiosperm seed contains a single embryo, derived from a single fertilisation event.

The Exception: In polyembryony, a single seed contains more than one embryo, often because additional embryos develop from nucellar tissue surrounding the embryo sac (independent of fertilisation).

Why (Reasoning): This phenomenon is agriculturally useful, since nucellar embryos are genetically identical to the mother plant, allowing propagation of true-to-type, disease-resistant seedlings.

NCERT / Classic Example: Polyembryony is common in Citrus species.

4. Seeds that sprout before they even leave the parent

General Rule: Seeds normally undergo a dormant period and germinate only after dispersal from the parent plant, once suitable external conditions are met.

The Exception: In vivipary, seeds germinate while the fruit is still attached to the parent plant.

Why (Reasoning): In mangroves growing in unstable, waterlogged, saline mudflats, this early germination gives the seedling a head start and a better chance of quickly anchoring itself once it eventually drops.

NCERT / Classic Example: Rhizophora (mangrove) shows vivipary — seedlings develop and drop as ready-rooted propagules.

5. A flower that never opens

careers360

General Rule: Flowers are generally understood to open their petals to expose reproductive parts, facilitating cross-pollination by wind, water or pollinators.

The Exception: Cleistogamous flowers remain permanently closed and are therefore obligately self-pollinated, since pollen from the anther has no way to leave (or enter from another flower).

Why (Reasoning): Because these flowers never open, cleistogamy guarantees self-pollination and seed-set even when pollinators are scarce, though it sacrifices the genetic benefits of cross-pollination.

NCERT / Classic Example: Commelina and Viola are cited in NCERT as producing both open (chasmogamous) and closed (cleistogamous) flowers.

6. Two fertilisation events in one embryo sac

General Rule: In most sexually reproducing organisms, a single fertilisation event (fusion of one sperm with one egg) is sufficient to initiate the formation of the next generation.

The Exception: Angiosperms undergo double fertilisation — one male gamete fuses with the egg cell to form the diploid zygote, while a second male gamete fuses with the two polar nuclei to form the triploid primary endosperm nucleus.

Why (Reasoning): This unique process, found only in flowering plants, simultaneously produces both the embryo and a ready-made nutritive tissue (endosperm) to nourish it — a key evolutionary innovation of angiosperms not seen in gymnosperms.

NCERT / Classic Example: Double fertilisation is a defining, unique feature of angiosperm reproduction (first described by S. G. Nawaschin).

careers360

Chapter 8: Genetics and Evolution in Plants

Plant genetics offers several classic deviations from straightforward Mendelian inheritance, along with a well-known example of convergent evolution.

1. Male sterility inherited only from the mother

General Rule: Most inherited plant traits are governed by nuclear genes, following the standard Mendelian pattern of inheritance from both parents.

The Exception: Cytoplasmic male sterility (CMS), where a plant fails to produce viable pollen, is controlled by genes located in the mitochondrial (cytoplasmic) DNA and is therefore inherited only through the maternal line.

Why (Reasoning): Because CMS is passed down only via the egg cell's cytoplasm (never through pollen), it does not follow standard Mendelian ratios and is instead exploited commercially to produce hybrid seeds without the labour of manual emasculation.

NCERT / Classic Example: CMS lines are widely used in hybrid seed production in crops.

2. The father's pollen changing this generation's seed

General Rule: The genetic contribution of the male parent (pollen) is normally expected to show up in the traits of the next generation (the embryo/plant that grows from the seed), not in the seed's own tissue in the current generation.

The Exception: In the xenia effect, the genotype of the pollen directly influences the phenotype of the triploid endosperm of the very same seed being formed — a visible effect in the current generation itself.

Why (Reasoning): Because the endosperm itself is formed by fusion with a second male gamete (see double fertilisation), traits like kernel colour or texture can immediately reflect the pollen parent's genes.

NCERT / Classic Example: Xenia effect on maize (corn) kernel colour and texture.

3. Genes that refuse to assort independently

General Rule: Mendel's Law of Independent Assortment states that genes for different traits are inherited independently of one another.

The Exception: In sweet pea, the genes controlling flower colour and pollen shape do not assort independently — they show linkage, tending to be inherited together because they lie close to one another on the same chromosome.

Why (Reasoning): This deviation from Mendel's expected ratios, first reported by Bateson and Punnett, laid important groundwork for the later chromosomal theory of linkage.

NCERT / Classic Example: Bateson and Punnett's classic sweet pea (Lathyrus odoratus) cross showing linkage between flower colour and pollen shape.

4. Two unrelated plants evolving the same look

General Rule: Similar-looking structures in different organisms are usually assumed to indicate a shared evolutionary origin (homology).

The Exception: Cacti (family Cactaceae) and many Euphorbia species belong to entirely different plant families, yet both have independently evolved fleshy, water-storing, spine-bearing stems as an adaptation to arid habitats.

Why (Reasoning): Because these similar structures arose independently in unrelated lineages facing the same environmental pressure (water scarcity), they are analogous structures produced by convergent evolution, not homologous ones.

careers360

NCERT / Classic Example: Cactus (Cactaceae) versus Euphorbia (Euphorbiaceae) — a textbook case of convergent evolution.

careers360

Chapter 9: Plants and Microbes in Human Welfare

Not all nitrogen-fixation or fermentation stories involve the organisms you'd expect — this chapter is full of useful exceptions.

1. Nitrogen fixation without a legume

General Rule: Biological nitrogen fixation via root-nodule symbiosis is most strongly associated with Rhizobium bacteria living in the roots of leguminous plants.

The Exception: The actinomycete Frankia forms a similar nitrogen-fixing, root-nodule symbiosis with a range of non-leguminous plants.

Why (Reasoning): This shows that the capacity for symbiotic nitrogen fixation is not restricted to the legume-Rhizobium partnership, but has evolved independently in other plant-microbe systems.

NCERT / Classic Example: Frankia in the root nodules of Alnus (alder).

2. A photosynthetic organism that also fixes nitrogen

General Rule: Nitrogen-fixing organisms (like Rhizobium) are typically non-photosynthetic bacteria that depend on their host plant for food.

The Exception: Cyanobacteria such as Anabaena and Nostoc are photosynthetic (oxygen-evolving) organisms that are also capable of fixing atmospheric nitrogen, usually within specialised cells called heterocysts.

Why (Reasoning): Combining autotrophy with nitrogen fixation makes cyanobacteria valuable free-living or symbiotic biofertilisers, as they do not need to derive their carbon/energy from a host.

NCERT / Classic Example: The Azolla-Anabaena symbiotic association is used as a biofertiliser in rice paddy fields.

3. A fungus that makes alcohol

General Rule: Fermentation to produce alcoholic beverages might be assumed to require specialised bacteria.

The Exception: It is actually the fungus Saccharomyces cerevisiae (yeast) that is used industrially to ferment sugars into ethanol.

Why (Reasoning): Yeast's ability to survive and ferment sugars anaerobically makes it uniquely suited for both baking (CO_2 for leavening) and alcoholic fermentation.

NCERT / Classic Example: Saccharomyces cerevisiae used in brewing, wine-making and bread-making.

4. A fungus that helps a root absorb more, not less

General Rule: Fungi are often thought of as parasites that harm their plant hosts.

The Exception: Mycorrhiza is a mutualistic association between certain fungi and plant roots, in which the fungus significantly improves the plant's absorption of phosphorus and other minerals from the soil.

Why (Reasoning): In exchange for carbohydrates from the plant, the fungal hyphae — with their much larger surface area — extend the effective absorptive reach of the root system, benefiting both partners.

NCERT / Classic Example: Mycorrhizal association in Pinus (its seeds cannot germinate without the appropriate mycorrhizal fungus present).

Careers360

Chapter 10: Biotechnology and Its Applications in Plants

Genetic engineering has produced plants that do things no naturally-evolved plant would.

1. A plant that manufactures its own insecticide

General Rule: Plants generally rely on external application of pesticides, natural resistance, or physical/chemical defences evolved on their own to fend off insect pests.

The Exception: Bt cotton and Bt brinjal are transgenic plants engineered with a cry gene from the soil bacterium Bacillus thuringiensis, causing the plant tissue itself to produce a protein toxic to specific insect larvae.

Why (Reasoning): Because the Bt toxin is only activated in the alkaline gut of a susceptible insect (and not in the acidic human gut), these crops are considered pest-resistant without being directly harmful to humans who consume them.

NCERT / Classic Example: Bt cotton — widely cultivated to resist the cotton bollworm.

2. A fruit engineered to stay firm longer

General Rule: Fruit ripening (and subsequent softening/spoilage) is a natural process typically triggered by the plant's own ethylene production.

The Exception: The Flavr Savr tomato was genetically engineered to slow down the production of an enzyme (polygalacturonase) responsible for softening, delaying over-ripening and spoilage.

Why (Reasoning): This was one of the earliest commercially significant applications of genetic engineering in crops, aimed at improving shelf life for transport and storage.

NCERT / Classic Example: Flavr Savr tomato — an early genetically modified crop developed for delayed ripening.

3. Rice that makes a nutrient it normally can't

General Rule: Rice endosperm, the edible part of the grain, normally does not produce beta-carotene (the precursor of Vitamin A).

The Exception: Golden Rice is a genetically engineered variety in which genes have been introduced to enable the endosperm to synthesise beta-carotene, giving the grain a golden-yellow colour.

Why (Reasoning): This was developed specifically to help combat Vitamin A deficiency in populations that rely heavily on rice as a dietary staple.

NCERT / Classic Example: Golden Rice — engineered to biosynthesise beta-carotene in the grain.

careers360

Chapter 11: Ecology and Environment — Plant Interactions

Plants also show some ecological relationships that break the usual expectations of self-sufficiency and independence.

1. A plant that eats insects

General Rule: Plants are autotrophic and normally obtain all the nitrogen and minerals they need from the soil through their roots.

The Exception: Insectivorous (carnivorous) plants trap and digest small insects to supplement their nutrition, particularly to obtain nitrogen, in habitats where the soil is nutrient-poor (such as bogs and marshes).

Why (Reasoning): Despite still photosynthesising normally, these plants have evolved specialised trapping structures to compensate for the lack of usable nitrogen in their native, mineral-deficient soils.

NCERT / Classic Example: Drosera (sundew), Nepenthes (pitcher plant) and Utricularia (bladderwort) are classic insectivorous plants.

2. A plant that lives entirely off another plant

General Rule: Ecological relationships between plants are often ones of simple competition for light, water and nutrients, with each plant otherwise self-sufficient.

The Exception: Cuscuta is a total parasite — having no chlorophyll or proper roots of its own — that entirely depends on a host plant for water, minerals and food, connected via specialised absorptive structures called haustoria.

Why (Reasoning): This total dependence goes well beyond ordinary competition, making Cuscuta a true parasite rather than a competitor.

NCERT / Classic Example: Cuscuta growing coiled around a host hedge plant, visibly leafless and rootless.

3. A partnership so specific that neither partner can survive without the other

General Rule: Most plant-pollinator relationships are relatively generalised — a given flower may be visited by several different pollinator species, and vice versa.

The Exception: Fig trees (Ficus) and fig wasps show an extremely specific, obligate mutualism — a particular fig species is pollinated by one particular wasp species, and that wasp can only reproduce inside that fig's inflorescence.

Why (Reasoning): This tight coevolutionary relationship means neither partner can complete its life cycle without the other, making it one of the most specific mutualisms known in the plant kingdom.

NCERT / Classic Example: Ficus (fig) and its species-specific pollinating wasp.

careers360

Other Useful Resources

Continue your NEET preparation with these free, exam-focused resources — sample papers, PYQs, mock tests, formula sheets and more, all available at careers360.com.

1. NEET Biology Sample Paper 2027 (Latest Pattern + Full-Length Practice PDF)
2. Top 100 NEET Physics Questions 2027 (Concept-Based & Numerical) – PDF with Solutions
3. Top 100 NCERT-Based Questions for NEET 2027 (Physics, Chemistry & Biology) – Most Expected PDF
4. Top 100 NEET Biology Questions to Score 330+ Marks (PYQ-Based + Concept-Focused PDF)
5. NEET Biology NCERT Line-by-Line Questions PDF – FREE Download with Answers & Detailed Solutions
6. NEET Inorganic Chemistry: 25 Must-Do Questions with Solutions PDF (Free Download)
7. NEET 2027 Magnetism & Matter PYQs PDF (50 Questions) – FREE Mock Test with Detailed Solutions
8. Chemical Bonding NEET Previous Year Questions with Solutions PDF – Must-Do Questions (Free Download)
9. Plant Growth and Development NEET Previous Year Questions PDF – 50+ Must-Do Questions (Free)
10. Top 20 Multi-Concept Questions for NEET Physics 2027 with Detailed Solutions (PDF)
11. Top 75 Most Repeated NEET Chemistry Questions for 2027 (PYQ-Based) – Free PDF
12. Electromagnetic Waves MCQs for NEET 2027 (Concept-Based Practice PDF)
13. NEET vs NCERT Syllabus 2027: Common Chapters, Deleted Topics & Study Map (Free PDF)
14. Top 50 Must-Practice NEET Physics Numericals with Solutions – Free PDF to Target 120+ Marks
15. Biology Mind Maps For NEET 2027 – Ebook Download
16. NEET 2027 Human Physiology PYQs – Ebook Download
17. NEET Mock Test Chemistry 2027 – Ebook Download
18. Most Scoring Concepts for NEET (As Per Latest NTA Syllabus) – Ebook Download
19. NEET Previous 10 Years Questions with Detailed Solutions – Ebook Download
20. Botany Mock Test for NEET 2027 – Ebook Download
21. Botany for NEET Exam – High-Scoring Concepts
22. NEET 2025 Question Paper with Answer Key (All Sets)
23. NEET 2025 Question Paper – Code 47 with Answer Key
24. NEET 2025 Question Paper – Code 48 with Answer Key & Solutions
25. NEET 2025 Question Papers with Detailed Solutions (Code 45, 46, 47 & 48)
26. NEET 2025 Subject-wise Analysis
27. NEET 2025 Question Papers with Answer Keys (All Sets)
28. NEET UG Previous Year Papers (Codes 45, 46, 47, 48)
29. NEET 2025 Code 45 Question Paper with Solutions
30. NEET 2025 Code 46 Question Paper with Solutions
31. NEET 2027 Formula Sheet – Physics & Chemistry
32. NEET Important Formulas – Chemistry
33. NEET Important Formulas – Physics
34. NEET Class 11 Biology Important Diagrams PDF

careers360

35. NEET Chemistry Cheat Sheet – Must-Know Formulas for Quick Revision
36. NEET Allen Scholarship Test – Preparation Kit (Class 11)
37. NEET Unacademy Scholarship Test – Preparation Kit (Class 11)
38. NEET Aakash National Talent Hunt Exam – Preparation Kit (Class 11)
39. NEET Physics Wallah Scholarship Test – Preparation Kit (Class 11)
40. NEET Narayana Scholarship Test – Preparation Kit (Class 11)
41. NEET 2024 Paper Analysis & Answer Key
42. How to Prepare for NEET with Class 12 Board Exam
43. NEET Biology NCERT-based Mind Maps & Diagrams
44. NEET Genetics & Evolution PYQs (100 Questions with Solutions)
45. NEET Biotechnology & Its Applications PYQs (100 Questions with Solutions)
46. NEET Ultimate 60-Day Study Plan – Crack NEET in 2 Months
47. NEET Most Scoring Chapters & Topics (Past 5-Year Analysis)
48. NEET 2015 Official Question Paper with Detailed Solutions
49. NEET 2016 Official Question Paper with Detailed Solutions
50. NEET 2017 Official Question Paper with Detailed Solutions
51. NEET 2018 Official Question Paper with Detailed Solutions
52. NEET 2019 Official Question Paper with Detailed Solutions
53. NEET 2020 Official Question Paper with Detailed Solutions
54. NEET 2021 Official Question Paper with Detailed Solutions
55. NEET 2022 Official Question Paper with Detailed Solutions
56. NEET 2023 Official Question Paper with Detailed Solutions
57. NEET 2024 Official Question Paper with Detailed Solutions
58. NEET Previous 10 Years Papers with Solutions
59. NEET Mock Test – Biology
60. NEET Mock Test – Physics
61. NEET Mock Test – Chemistry
62. NEET Botany Mock Test 2027
63. NEET Zoology Mock Test 2027
64. Biology Comparison & Differences Between Concepts – NEET 2027

Careers360